

# AN AI POWERED SOFTWARE-BASED FIRE DETECTION SYSTEM FOR CCTV PLATFORMS.

## FiremeX - TEST PLAN

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|--------------------|---------------------|
| <b>Document ID</b> | QA - 02             |
| <b>Project</b>     | FiremeX             |
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## 1. Objectives

- Verify fire and smoke detection accuracy under varied lighting, weather, and camera conditions.
- Validate real-time alert generation and notification delivery within defined SLA ( $\leq 5$  seconds).
- Confirm integration with camera feeds (ONVIF/RTSP), dashboard UI.
- Ensure system reliability for 24x7 continuous monitoring with minimal false positives/negatives.
- Validate data storage, audit logging, and historical incident retrieval.
- The system uses historical alerts and incident records to identify recurring patterns and continuously improve the AI model, reducing false alarms and increasing detection accuracy for frequently occurring scenarios.

## 2. Scope

### 2.1 In Scope

- Functional testing of video ingestion, detection engine, alerting, and dashboard modules.
- Non-functional testing: performance, load, security, and reliability of the detection pipeline.
- Integration testing with CCTV/NVR systems, notification gateways (SMS/Call).
- Regression testing for every AI model version update.
- Usability testing of the monitoring dashboard for control-room operators.

### 2.2 Out of Scope

- Testing of third-party fire-suppression hardware actuation (mechanical).
- Detailed AI model training pipeline validation (covered under Data Science validation plan).

## 3. Test Approach / Methodology

| Test Level          | Description                                                               | Owner      |
|---------------------|---------------------------------------------------------------------------|------------|
| Unit Testing        | Developer-level testing of detection algorithm modules and APIs           | QA Members |
| Integration Testing | Camera feed ingestion, AI engine, alert & notification module integration | QA Members |
| System Testing      | End-to-end validation of fire detection to alert delivery flow            | QA Members |
| Performance Testing | Multi-camera concurrent stream load and detection latency testing         | QA Members |
| Security Testing    | Access control, API vulnerability testing                                 | QA Members |
| UAT                 | Control room operators validate real-world usability and alert clarity    | End Users  |
| Regression Testing  | Re-validation after every model & software build update                   | QA Members |

## 4. Test Environment

- Staging environment mirroring production: IP CCTV cameras (indoor/outdoor), NVR, GPU inference server.
- Simulated fire/smoke video datasets (day, night, fog, rain) for regression suites.
- Test dashboard instance connected to a sandbox SMS/Call gateway and mock fire-panel API.

## 5. Entry Criteria

- Build deployed successfully to QA/staging environment.
- Unit testing completed with  $\geq 90\%$  code coverage on core detection modules.
- Test data sets (fire/smoke/no-fire video clips) prepared and validated.
- Test cases reviewed and approved.

## 6. Exit Criteria

- 100% of planned test cases executed;  $\geq 95\%$  pass rate.
- No open Critical or High severity defects.
- Detection accuracy  $\geq$  agreed threshold (e.g., 95% true positive,  $\leq 2\%$  false positive on validation set).
- Performance and security sign-off obtained.

## 7. Suspension & Resumption Criteria

Testing will be suspended if the detection engine fails to start, camera feed ingestion is completely broken, or a Critical defect blocks more than 30% of planned test cases. Testing resumes once a fix is deployed and smoke-tested.

## 8. Roles and Responsibilities

| Role                 | Responsibility                                           |
|----------------------|----------------------------------------------------------|
| QA Lead              | Test planning, strategy, sign-off, stakeholder reporting |
| Test Engineers       | Test case design, execution, defect logging              |
| Automation Engineer  | Regression automation suite for detection & UI           |
| Performance Engineer | Load/stress testing of video pipeline                    |
| Business Analyst     | Requirement clarification, UAT coordination              |
| Developers           | Defect fixes, unit testing, environment support          |

## 9. Test Deliverables

- Test Plan, Test Cases, Traceability Matrix
- Test Execution Report, Defect Reports
- Performance & Security Test Reports
- Test Closure Report

## 10. Schedule

| Phase                          | Start   | End     |
|--------------------------------|---------|---------|
| Test Planning & Design         | Week 06 | Week 09 |
| Test Environment Setup         | Week 10 | Week 11 |
| Functional Testing             | Week 12 | Week 21 |
| Integration Testing            | Week 18 | Week 21 |
| Performance & Security Testing | Week 21 | Week 24 |
| UAT                            | Week 25 | Week 25 |
| Test Closure & Sign-off        | Week 27 | Week 28 |

## 11. Risks and Mitigation

| Risk                                                   | Mitigation                                                                  |
|--------------------------------------------------------|-----------------------------------------------------------------------------|
| Insufficient real fire/smoke video samples for testing | Use synthetic/augmented datasets and controlled test burns from vendor labs |
| High false-positive rate impacting reliability         | Threshold tuning cycles with QA feedback before each release                |
| Network latency affecting real-time alerting           | Dedicated performance test on production-like network conditions            |
| Camera hardware variability (brand/resolution)         | Test matrix covering multiple camera makes/models supported                 |